

VisaudiBlow: Fine-grained Blowing Interaction Based on Visual and Auditory Detection

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ABSTRACT

More and more studies regard blowing as a means of reflecting the user's physiological conditions. Its potential as an independent input channel has not been fully explored. In this paper, we analyze the design space of blowing as an independent interactive channel and propose a novel blowing recognition approach, VisaudiBlow, based on visual and audio detection. VisaudiBlow uses Facial Tracker to capture the mouth expression when doing specific blowing actions, and audio information generated by blowing is obtained by a microphone beside the mouth, enabling identification and detection of the input parameters such as blowing type, direction, and intensity. In order to verify the validity of blowing as an independent input channel, a prototype system BlowUI is implemented. The experimental results demonstrate that blow interaction can effectively complete the menu control tasks and achieve the same interaction efficiency as controllers in specific scenarios.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**; *Virtual reality*.

KEYWORDS

BlowUI, mouth expression, blowing interaction, blowing parameters

*Both authors contributed equally to this research.

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1 INTRODUCTION

Blowing is one of the most natural physiological activities of human beings and is also considered as an alternative control mechanism that affects the physical world and the virtual environment [24]. Existing VR applications mainly focused on eye-movement interaction [14, 19], gesture interaction [18], voice interaction [10]. Only a few works use breathing as a response of physiological state or user experience in VR games. Other works use specific blowing action as an auxiliary interaction channel for input methods, such as blowing as a trigger mechanism for gestures [27]. We are concerned about the many people in the world with arm and hand disabilities who are unable to use traditional joystick interaction or gesture interaction during interaction. These people who are unable to use traditional joystick interaction or gesture interaction during interaction usually use their eyes or voice for basic interface interaction. But voice interaction is affected by noise, and eye-movement interaction contains little interaction information. Blowing can be a good solution to these problems, and we believe that these people can perform basic VR interface interaction by just blowing.

Through the movement around the mouth muscles, people can control the direction, velocity, duration, and other parameters of the airflow when blowing. The current research is mainly based on a certain type of sensor, such as microphones, vibration sensors, cameras, and so on, to obtain information such as the intensity of

blowing sounds, the frequency of blowing actions, and the orientation of the face. Because the direction of airflow when blowing is difficult to obtain through a single type of optical and vibration sensor, there is few relevant research.

By combining facial expression recognition technology with auditory recognition technology, the fine actions around the mouth and the acoustic characteristics of blowing can be detected, which leads to more elaborate blowing recognition, further increasing the design space of blowing as an interaction channel and improving its control ability.

We propose VisaudiBlow, a visual and auditory fusion blowing detection technology that combines facial and audio information when blowing. We use the Facial Tracker to capture the changes in mouth expression to determine the direction of blowing. And we use a microphone to capture the audio information of blowing to represent the intensity of blowing. Based on these detected parameters, we designed two blowing interaction-based applications, which proved the feasibility of blowing as an independent interaction channel. The main contributions of our work are as follows:

- We propose the possibility of taking blowing as an independent interaction channel and implementing a dual-channel blowing detection technology based on visual and auditory features - VisaudiBlow, which can identify the characteristic parameters of various blowing actions, including blowing intensity, blowing direction and inhalation.
- We designed BlowUI, a user interface system based on blowing interaction. It utilizes a variety of blowing parameters to realize the interactive control of the typical interfaces in VR.
- Based on BlowUI, we conducted two user studies to investigate the performance of the blowing parameters and to verify the feasibility of using blow as an independent interaction channel.

2 RELATED WORK

2.1 Breathing as physiological parameters in VR

Research on blowing interaction in VR can be divided into two types. Most work treats blow as an indicator of the physiological status, while a few use breath as a factor affecting the environment. Rooij et al. [26] used VR games to treat anxiety disorders of children. They monitored the children's breath rate and other physiological information to help judge the effects of the treatment. Adler et al. [1] studied the Visuo-respiratory conflict in VR and revealed that Visuo-respiratory conflicts would alter body self-consciousness and breathing agency.

Simulating the effect of breathing on a real environment in a VR setting can enhance the fun or realistic experience of a game. For example, in the Inner Garden VR-assisted mindfulness system developed by Roo et al. [21], users can raise or lower the sea level in the scene by breathing. Soyka et al. [23] developed a VR underwater world that allows users to adjust their breathing rhythm following the fluctuations of jellyfish. Kors et al. [9] designed a mixed-reality game called ABTJ. ABTJ places the player in the shoes of a refugee who is fleeing from a war-torn country. The system would determine whether the protagonist is able to avoid detection or not with a breath sensor.

2.2 Application of blowing parameters in interaction studies

Existing researches on blowing interaction focus on blowing intensity [20, 24], duration [17], frequency [6, 7, 11, 13], etc. Sra et al. [24] divided blowing into five types according to duration, intensity and frequency, and explored the application of these five blowing modes in games. Marshall et al. [15] use breathing to control the rotation of a riding device. Inhale to rotate clockwise and exhale to rotate anti-clockwise. The speed of rotation is modulated by the speed of blowing.

There is also work that uses the direction of blowing. For example, Kim and Lee [8] calculated the direction of blowing by the position and angle between the user and the mobile device. Patel and Abowd [16] used a microphone on a laptop computer to identify where the user was blowing onto screen. However, the judgement of the blowing direction in the above work is based on the rotation of the head, while the judgement of blowing direction based on the mouth expression has not been studied.

2.3 Detection means of blowing parameters

Most traditional blowing recognition methods are based on vibration sensors to identify blowing parameters such as the intensity and duration. Kuzume et al. [12] used a piezo film sensor array to detect the exhalation signal and a bone conduction microphone to detect the sound signal from tooth-touch contact, thus realizing a hands-free input device for the disabled. Desnoyers-Stewart et al. [3] used a Biosignalsplux piezoelectric breathing sensor around the diaphragm to detect the rise and fall of the abdomen, so as to control the movement of the virtual jellyfish. Sra et al. [24] identified blowing parameters by wearing a respiratory detection belt, the Zephyr Biostrap. Markus Tatzgern et al. [25] enhanced the virtual experience with a self-developed AirRes mask and could potentially create more immersive scenarios for applications by enhancing the perception of danger or increasing situational awareness in training simulations. Corbishley et al. [2] proposed a miniature blowing detection system that uses a microphone placed in the neck to listen for acoustic signal of airflow. Georgia Institute of Technology [5] developed a RVSM (Rader sensing of heartbeat and respiration) system that could monitor human heartbeat and respiration signals at a distance of more than 10m.

Due to the small movement of shoulders when breathing, Shao et al. [22] explored an optical detection approach based on camera. They analysed the movement of a small area of the shoulder image over time, and monitored vital physiological signals such as heart rate, pulse time and breathing patterns.

Visual information allows for recognition of mouth movements, resulting in more precise blowing directions, but there is no blowing detection based on facial expression recognition. However, when using the head-based orientation, if the user's head cannot move, then the traditional blowing interaction system cannot be used. And since VR users have a lot of directional interaction needs, we propose BlowUI to supplement the lexical space with blow direction and provide quick operation.

3 VISAUDIBLOW

In this session, we will introduce in detail VisaudiBlow, a dual-channel blowing detection approach that is used to identify the direction and intensity of blowing and detect the user's inhalation. We use a lavalier microphone to receive audio information and HTC Vive Facial Tracker to detect changes in facial expression, which can capture the facial actions through 38 tracking points on the lips, chin, teeth, tongue, and cheeks.

3.1 Determination of blowing and inhaling

Most existing works rely on audio information to determine whether users are blowing or inhaling, but the sound of inhalation is relatively small, which increases the difficulty of detection. In VisaudiBlow, we use cheek depression as a basis for determination. In this way, it can be judged even in a noisy background, so as to obtain better robustness.

However, facial changes may not necessarily be accompanied by actual blowing actions. As in previous work, considering that the sound is usually emitted during inhaling or blowing, we also detected the occurrence of blowing actions based on changes in the volume of the microphone.

3.2 Recognition of blowing direction

Most current studies on blowing direction use the front of the mouth as the fixed direction. This is actually replacing the direction of blowing with the direction of the head, which we called head-based blowing direction. Based on this approach, interactive operations need to be completed by turning the position of the head several times in VR. And it is only applicable to the scene where the interactive objects are fixed in the VR environment and do not move with the head.

In this paper, the recognition of blowing direction is based on visual features, which can realize a finer-grained recognition of blowing direction, thus greatly increasing the lexical space of blowing interaction. We call it expression-based blowing direction.

As shown in Fig. 1, when the user blows in different directions, Facial Tracker will return the values of the corresponding tracking points on the user's face. For example, When determining the user to blow to the left, we need to focus on the facial tracker to get the Mouth-Upper-Left and Mouth-Lower-Left two parameters, other parameters such as Jaw-Open and Cheek-Puff-Right can be used as an auxiliary reference

3.3 Detection of blowing intensity

Blowing intensity can naturally be used as a continuous interactive analog. We use a lavalier microphone to receive volume information when the user is blowing or inhaling. The volume information not only determines whether the user is blowing or inhaling at that moment, but can also be mapped to blowing intensity. In BlowUI, changes in blowing intensity are employed to control the speed of menu sliding.

4 BLOWUI

Previous studies on blowing interaction mainly utilized one parameter of blowing as an auxiliary input channel but did not fully explore the ability of blowing as an independent interaction channel. We

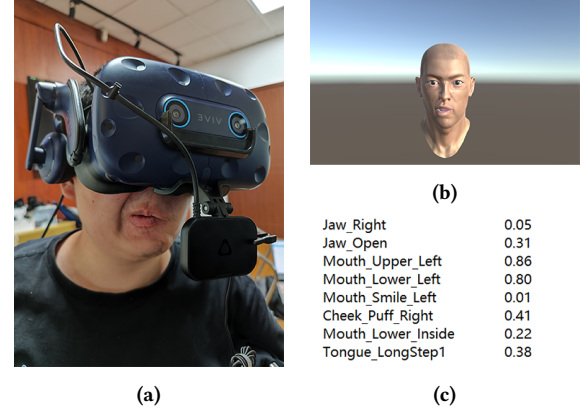


Figure 1: Recognition of blowing direction. (a) Expression when blowing to the left. (b) Facial Tracker recognized. (c) Expression parameters captured by facial tracker.

believe that we can construct a blowing interaction channel with powerful interaction performance by utilizing parameters such as blowing direction, intensity, duration, etc.

The blowing direction component can be mapped to direction-related actions of the user interface like selection actions of the pie menu. The blowing intensity component can be mapped to the speed control component of UI operations, such as the sliding speed of the sliding menu. Blowing duration can be used to zoom in or zoom out the interface elements. Exhalation can be used as a trigger switch for certain interface input states, and inhalation can be used to undo the previous operation, and so on.

We designed a prototype system BlowUI, which uses blowing as the only interaction input channel. Here, the user can perform interface operations, like menu item selection, just by blowing. The blow recognition process is shown in Fig. 2. BlowUI first detects the user whether blows or inhale. When inhaling, inhaling intensity will be identified at the same time. When blowing, BlowUI also determines the direction and intensity of the blow. The system interface design will be discussed in detail in the experiment section.

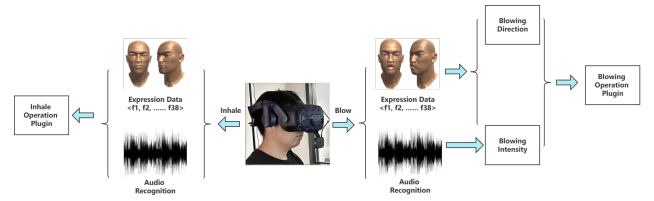


Figure 2: The detection process of BlowUI.

5 USER STUDY

In this section, we validated the blowing detection technique based on facial and auditory detection and explored the ability of utilizing BlowUI as an input modality in VR applications. In particular, the ability to apply blowing direction, intensity and inhalation to menu control and interact with different types of menus.

5.1 Apparatus and Participants

The experiment employs an HTC Vive Pro Eye HMD as the VR kit. Facial expression recognition is based on the HTC Vive Facial Tracker. And we used a Sudotack Lavalier microphone to capture the sound of blowing.

The participants were 12 college students on campus. They had an average of 0.8 years of experience with VR devices, ranging from 0 to 4 years, but had no experience with blowing interaction in VR.

5.2 Experiment Design

We set up two commonly used menu forms, i.e., pie menu and sliding menu, to explore the control ability of blowing interaction on different types of menus. We compared the blowing control on the pie menu with the control of using controllers, which is user study1, and compared which blow mode is better for controlling sliding menus, which is user study2.

In summary, there are four experiment conditions: the blowing interaction group and the traditional controller interaction group in user study1, and the experimental groups with two blowing modes in user study2. We randomly divided the 12 participants equally into four groups, each of which was required to complete all four experiment condition tasks, but the order of the conditions differed between each group. The condition order was randomly balanced among the four groups using a Latin square.

Since each participant had a different mouth shape and amplitude when blowing. Therefore, before the experiment began, each participant was required to wear the device and blow in each direction ten times in turn. Based on the facial data and audio data, the "threshold" of each participant's blowing in each direction was calculated and saved as personal data. When the mouth shape and amplitude of the blow and the volume of the blow are within the "threshold", the participant is judged to be blowing at that moment.

5.2.1 Pre-Study. Before User Study 1, a simple pre-study was conducted to verify the performance of the proposed detection approach in determining the blowing direction.

In pre-study, participants were asked to hit eight targets in different directions (i.e. Up, Down, Left, Right, Top-left, Top-right, Bottom-left, and Bottom-right) by blowing as quickly and accurately as possible. The target in each direction would appear in a random manner and only once. The number of blows and hits of each task would be recorded.

Fig. 3(left) shows the probability of the participants hitting the target by blowing. It can be found that the hit rate of the four directions, Up, Down, Left and Right, is much higher than in the remaining four directions. Fig. 3(right) shows the average times of blowing in each direction. It can be seen that the times of blowing in the left and right directions is the least, which only needs about 1.5 times on average. Both the Up and Down directions require a second chance. The remaining four directions require more than a second chance for almost every shot. This shows that it is easier to do the standard mouth shape when blowing in four directions: up, down, left and right.

This may be because people usually do not blow in the oblique direction daily, so it is difficult to do the oblique blowing action, which leads to inaccurate facial expression recognition. Therefore, in the

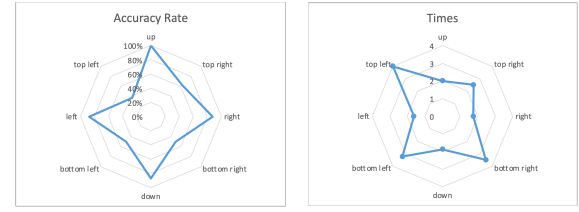


Figure 3: Accuracy rate(left) and times of blowing(right) in each direction in pre-study

design of the pie menu in study1, we only chose four directions: Up, Down, Left and Right.

5.2.2 User study1: blowing direction + inhalation in pie menu. This experiment is to verify the ability of blowing interaction to control the pie menus. The menu selection is a basic input task in VR applications. Due to the unique directional characteristics of blowing interaction, the pie menu is very suitable for it.

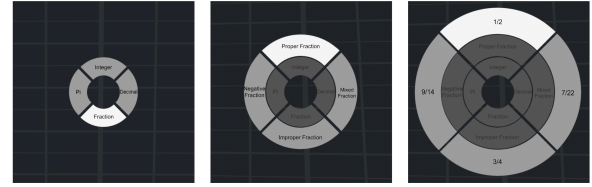


Figure 4: Choose "1/2" in the pie menu

Based on pre-study, the pie menu is designed into three layers, each with four sub-menus. The menu content is designed to be number classification tasks familiar to science and engineering students, as shown in Fig. 4. Concerning blowing interaction, the participants trigger the next sub-menu by blowing in different directions. The target may appear in any of the three layers. The task is deemed to be successful if the target menu is selected by blowing. If the selection is incorrect, it can be canceled by the inhalation action to return to the previous menu. Each participant had to complete eight menu selection tasks using both blowing interaction and a controller. While using the controller, participants aim at the target menu with a ray fired from the top of the controller, then pull the trigger button for menu selection and undo it with a side button on the controller. The time participants completed the task, and the number of invalid interactions were recorded. The experimental process is balanced with Latin square matrix.

5.2.3 User study2: blowing intensity in sliding menu. In user study2, we designed two ways to interact with the sliding menu based on blowing, namely Gale and Gust [24]. Gale refers to a strong, sustained blowing pattern similar to cake blowing. Gust means a transient but strong jet of air blown from the mouth (similar to blowing the dust away from objects).

In Gale mode, the speed of menu sliding is proportional to the intensity of blowing. The more intense the blow, the faster the menu will slide. Once the blowing action stops, the menu also stops sliding. In Gust mode, the initial movement speed of the menu depends on the strength of a Gust and then slows down until it

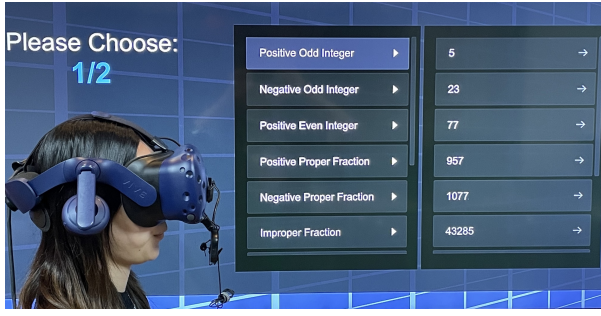


Figure 5: Using blowing to manipulate the sliding menu

stops. Here, the sliding menu is designed in two layers. There are 11 menu items in the first layer, and each menu item corresponds to 10 second-layer menus (Fig. 5). Similar to study1, the user can slide the menu up and down, back to the previous layer, or to the next layer by blowing in four directions (UDLR). Each participant was randomly assigned five menu-selected tasks.

5.3 Results and Discussion

Since the task of menu selection through blowing interaction was too unfamiliar for most participants, and we also wanted to analyze the improvement of blowing interaction performance with the improvement of proficiency, we performed each experiment twice (1st experiment and 2nd experiment) with an interval of two days. There was no practice between the two experiments. In the two experiments, the average time used to complete the menu selection task was about 20 minutes ($m_1 = 20.67min, \sigma_1 = 9.60min$) ($m_2 = 17.75min, \sigma_2 = 8.08min$). It can be considered that the first experiment reflects the performance of the first experience of blowing interaction, and the second experiment is the performance after relatively proficient. We used Paired t-tests ($P < .05$) to analyze the average completion time in both experiments and the number of menu slides in Experiment 2. Wilcoxon signed-rank test ($P < .05$) was employed to analyze the error rate in user study1.

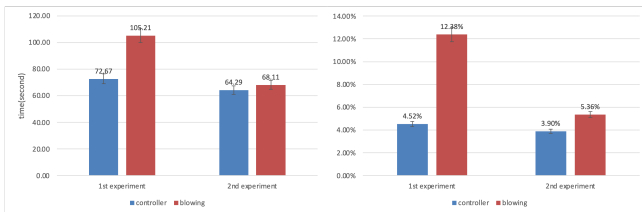


Figure 6: Quantitative results of user study1: average task completion time (left) and average error rate (right).

Fig. 6 shows the data of user study1. In the 1st experiment, blowing interaction ($m = 105.21s, \sigma = 30.74s$) and controller interaction ($m = 72.67s, \sigma = 14.61s$) had significant differences in average task completion time ($T(11) = 4.74, P < .005$). Similarly, the error rate with blowing ($m = 12.38\%$) was significantly higher than that with controller ($m = 4.52\%$) ($P < .05$). In the second experiment, blowing significantly reduced task completion time

($m = 68.11s, \sigma = 9.70s$) and error rate ($m = 5.36\%$). The completion time ($m = 64.29s, \sigma = 10.09s$) and error rate ($m = 3.90\%$) with controller are almost the same, and the difference in time ($T(11) = 2.10, P > .05$) and error rate ($P > .05$) is not significant.

It can be seen that the blowing interaction in this paper is comparable to the traditional operation method after proficiency, both in terms of efficiency and accuracy. This increase in efficiency and accuracy after only one more experiment and two days apart shows that pie menu selection is well mastered using blowing interactions.

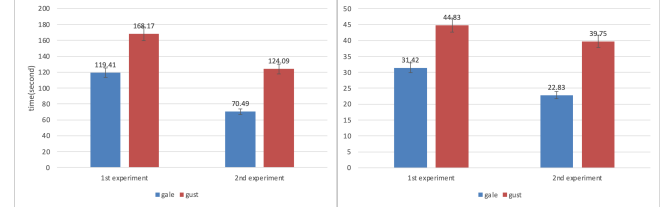


Figure 7: Quantitative results of user study2: average task completion time (left) and average times of menu slides (right).

Fig. 7 demonstrates the average completion time and the average number of menu bar sliding up and down in user study2. It can be found that under the two different sliding modes, the two experiments both have obvious changes, which is similar to user study1. Concerning task completion time, the Gale mode takes significantly ($T(11) = 5.012, P < .005$) shorter time ($m = 119.41s, \sigma = 52.07s$) than the Gust mode ($m = 168.17s, \sigma = 60.96s$). The number of slides using Gale mode ($m = 31.42, \sigma = 10.66$) was also significantly less than that using Gust mode ($m = 44.83, \sigma = 11.25$) ($T(11) = 5.90, P < .005$).

The task completion time ($m_{Gale} = 70.49s, \sigma = 16.70s$) ($m_{Gust} = 124.09s, \sigma = 26.60s$) and the number of slides ($m_{Gale} = 22.83, \sigma = 8.71$) ($m_{Gust} = 39.75, \sigma = 7.92$) of both modes decreased as they became more familiar with blowing interaction. However, there was a significant gap between them ($T_{Gale}(11) = 6.99, P < .005$) ($T_{Gust}(11) = 6.58, P < .005$). In summary, Gale performs better than Gust when interacting with sliding menus. This may be due to the fact that the stop of menu sliding is easier to control in Gale mode, rather than being as unpredictable as Gust to predict where the cursor will end up.

6 CONCLUSION AND FEATURE WORK

In this paper, we propose using blowing as an independent interactive channel and its available input parameters. And we implement a dual-channel blowing detection technology based on auditory and visual fusion, VisaudiBlow. In order to verify the feasibility of VisaudiBlow, we designed and implemented a VR prototype BlowUI and explored the feasibility and efficiency of using blowing direction, intensity, and inhalation to control the pie menu and sliding menu.

The experimental results suggest that blowing interaction can be used as an independent interaction input channel to complete the control of menus and can achieve the same efficiency as traditional controllers in the pie menu scenario. Concerning the sliding menu,

the Gale mode of real-time matching between menu sliding speed and blowing intensity has better performance.

Currently, the blowing detection approach in this paper is simple and primary. The "threshold" we set for each person to blow in each direction is simply a linear calculation, including facial data and audio data. And if a user could have made a loud sound like "ah" without really blowing, this would be incorrectly detected. In future work, we can combine head-based directions to provide a broader interactive lexical space, and it is believed that with the development of hardware devices and detection algorithms, blowing interaction will have a broader application prospect in VR.

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